

PRIVACY-PRESERVING FEDERATED FRAUD INTELLIGENCE

TrustFed

Product Requirements Document

TrustFed is a privacy-preserving federated learning platform that enables simulated financial institutions — a bank, a digital wallet, a lender and an insurer — to jointly train a fraud-risk model without ever exchanging raw customer or transaction data.

Product	TrustFed
Document	Product Requirements Document
Version	Hackathon MVP
Primary use case	Cross-institution fraud detection
Core message	Institutions collaborate on intelligence, not customer data.

REQUIREMENT NOTE

This document specifies a hackathon MVP. All institutions, datasets, privacy budgets and results described are simulated and are intended to demonstrate technical feasibility and measurable value, not production fraud prevention or regulatory certification.

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1. Product Overview

TrustFed is a privacy-preserving federated learning platform that enables simulated financial institutions—including a bank, digital wallet, lender and insurer—to jointly train a fraud-risk model without exchanging raw customer or transaction data.

Each institution keeps its dataset locally, trains a model on its own data, protects its model update through clipping, differential privacy and secure aggregation, and contributes only the protected update to a shared global model.

The platform provides:

- Cross-institution federated model training.
- Secure aggregation of model updates.
- Differential privacy controls and accounting.
- Explainable transaction-level risk decisions.
- Fairness and model-drift monitoring.
- A dashboard comparing local-only and federated performance.

REQUIREMENT NOTE

Flower provides the relevant federated-learning, secure-aggregation and differential-privacy workflows for the implementation.

2. Problem Statement

Financial institutions often possess complementary signals about the same fraud ecosystem:

Banks observe	account activity, transfers and balances.
Wallets observe	mobile-money transactions and device behaviour.
Lenders observe	repayment and borrowing patterns.
Insurers observe	claims and unusual behavioural patterns.

However, institutions cannot normally centralize raw customer data because of privacy, security, commercial and regulatory constraints.

As a result, each institution trains a siloed model with an incomplete view of fraud. TrustFed tests whether institutions can improve fraud detection collaboratively while keeping raw data inside institutional boundaries.

3. Goals and Non-Goals

Goals

1. Demonstrate that federated training can outperform silo-only models.
2. Ensure raw customer records never leave simulated institution clients.

3. Prevent the coordinator from inspecting individual client updates through secure aggregation.
4. Quantify privacy and utility trade-offs using differential privacy.
5. Provide explainable risk scores and recommended actions.
6. Monitor fairness across selected evaluation groups.
7. Detect model or feature drift over time.
8. Deliver a polished three-minute judge-facing demo.

Non-goals

- Production-grade fraud prevention.
- Legal certification or automatic regulatory compliance.
- Real customer data processing.
- Fully malicious-client detection.
- Production identity resolution across institutions.
- Automatic rejection of customers without human review.
- Solving fraud detection and credit default prediction simultaneously.

REQUIREMENT NOTE

Credit-risk modelling may be presented as a future extension, but fraud detection remains the sole MVP use case.

4. Target Users

User	Needs	Key actions
Institution administrator	Control over local participation and privacy settings	Configure data, join rounds, inspect local metrics
Risk analyst	Understand and act on transaction risk	Score transactions, review explanations, escalate cases
Compliance or governance user	Verify privacy, fairness and auditability	Inspect privacy scorecards, fairness metrics and logs
Hackathon judge	See measurable value and technical credibility	Compare model variants and observe the live demo

5. User Stories

Institution administrator

- As an institution administrator, I want to map my local dataset to a shared schema without uploading raw records.
- As an institution administrator, I want to set privacy parameters before joining a training round.
- As an institution administrator, I want to see my local model's performance versus the global model.
- As an institution administrator, I want to know exactly what information leaves my institution.

- As an institution administrator, I want to reject a training round if privacy or governance requirements are not met.

Risk analyst

- As a risk analyst, I want to submit a transaction for scoring.
- As a risk analyst, I want to see a fraud probability, risk category and recommended action.
- As a risk analyst, I want to understand the main factors contributing to a risk score.
- As a risk analyst, I want to compare local and federated predictions.
- As a risk analyst, I want high-risk transactions routed to manual review rather than automatically rejected.

Governance user

- As a governance user, I want to verify that no raw rows were shared.
- As a governance user, I want to inspect the configured privacy budget and minimum client threshold.
- As a governance user, I want to compare fairness metrics across customer groups.
- As a governance user, I want a complete audit trail for training and scoring actions.

6. MVP Scope

Included

- PaySim-based synthetic transaction dataset.
- Three simulated institutions: bank, wallet and lending app.
- Optional fourth institution: insurer.
- Non-IID data partitions.
- PyTorch fraud-classification model.
- Flower-based FedAvg training.
- Local-only, centralized, standard federated and privacy-enhanced federated baselines.
- Secure aggregation.
- Basic differential privacy.
- SHAP or equivalent local / per-transaction explanations.
- PR-AUC, recall, precision, F1, false-positive rate and ROC-AUC.
- Fairness metrics by selected evaluation groups.
- Basic feature / model drift monitoring.
- Streamlit or React dashboard.
- Audit log and privacy scorecard.
- Live training-round visualization.
- Human-review recommendations.

Deferred

- FedProx optimization.
- Private set intersection.

- Real multi-party identity matching.
- Production authentication and key management.
- Malicious-client defense.
- Blockchain.
- Full case-management workflow.
- Credit-default modelling.
- Production deployment across independent organizations.

7. Functional Requirements

FR-1: Institution onboarding

The system shall allow an institution administrator to:

- Select or upload a local dataset.
- Map local columns to the shared feature schema.
- View row count, fraud rate, missing values and feature statistics.
- Configure institution name and client identifier.
- Keep raw records within the local client environment.

REQUIREMENT NOTE

The coordinator shall receive only metadata explicitly approved for display, such as dataset size and aggregate fraud rate.

FR-2: Local preprocessing

Each client shall perform preprocessing locally, including:

- Missing-value handling.
- Categorical encoding.
- Numerical scaling where required.
- Identifier removal.
- Feature validation.
- Class-imbalance handling.
- Train, validation and time-based test splitting.

The system should prefer temporal splits over purely random row splits to better represent financial risk deployment.

FR-3: Local model training

Each institution shall train a local fraud model using its own data. The MVP model should be a small PyTorch multilayer perceptron:

```
graph TD; A[Input features] --> B[Dense layer: 64 units, ReLU]; B --> C[Dropout]; C --> D[Dense layer: 32 units, ReLU]; D --> E[Fraud probability output];
```

The training process shall support:

- Weighted binary cross-entropy or focal loss.
- Configurable local epochs.
- Configurable batch size.
- Reproducible random seeds.
- Local validation metrics.
- Model checkpointing.

FR-4: Federated training

The coordinator shall:

1. Initialize the global model.
2. Select participating clients.
3. Send the current global model to clients.
4. Receive protected client contributions.
5. Aggregate contributions.
6. Produce a new global model version.
7. Evaluate the model on federated or approved evaluation data.
8. Publish round-level metrics.

The MVP shall support FedAvg. FedProx may be added if non-IID data causes unstable convergence.

FR-5: Secure aggregation

The system shall use secure aggregation so that the coordinator cannot inspect any individual client update. The dashboard shall show:

- Number of participating clients.
- Number of updates received.
- Whether individual updates are visible to the coordinator.
- Whether an aggregate update was reconstructed.
- Minimum participation threshold.
- Client dropout status.

The system shall not finalize an aggregation round when the number of participating clients is below the configured threshold. This prevents a single client's contribution from becoming inferable.

REQUIREMENT NOTE

Flower provides SecAgg and SecAgg+ workflows designed to keep individual updates private while allowing aggregate updates to be computed.

FR-6: Differential privacy

The system shall support differential privacy through:

- Update or gradient clipping.
- Configurable clipping norm.
- Configurable noise multiplier.
- Privacy accounting.
- Cumulative privacy budget display.
- Configurable failure probability δ .

The dashboard shall display:

- DP status.
- Clipping norm.
- Noise multiplier.
- Number of rounds included in the privacy calculation.
- Privacy budget ϵ , when calculated by the selected accounting method.
- A clear indication when a value is a configured simulation parameter rather than a measured guarantee.

The system must distinguish between local DP and central DP. In local DP, protection is applied before an update leaves the client; in central DP, noise is applied to the aggregated result.

FR-7: Risk scoring

The risk analyst shall be able to submit a transaction manually or through a demo form. The system shall return:

- Fraud probability.
- Risk score from 0 to 100.
- Risk category: low, medium or high.
- Recommended action: approve, review or block.
- Model version.
- Model type: local or global.
- Explanation status.

Recommended MVP policy:

Risk score	Category	Recommended action
0–39	Low	Approve
40–69	Medium	Additional verification
70–100	High	Manual review

Thresholds must be configurable and tuned against validation results. The system shall not automatically deny a person in the demo.

FR-8: Explainability

For each scored transaction, the system shall show:

- Top contributing features.
- Direction of contribution.
- Feature value.
- Plain-language reason code.
- Model version.
- Explanation timestamp.

Risk score: 87/100

Decision: Review required

Main factors:

- Amount is substantially above the customer baseline.
- Destination account is new.
- Device has not been seen previously.
- Multiple transfers occurred within a short interval.

The MVP may use global feature importance plus SHAP-style per-transaction explanations.

FR-9: Fairness monitoring

The system shall evaluate model behaviour across available evaluation groups, such as:

- New versus existing customers.
- Thin-file versus thick-file customers.
- Customer tenure bands.
- Region.
- Income band, where available.

REQUIREMENT NOTE

Protected or sensitive attributes shall be isolated for evaluation and monitoring unless their use is explicitly justified.

The dashboard shall show:

- Recall by group.
- False-positive rate by group.
- False-negative rate by group.
- Precision by group.
- Equal opportunity difference.
- Demographic parity difference where appropriate.
- Sample size for every group.

The system shall warn users when a metric is unreliable because a group has too few examples.

REQUIREMENT NOTE

NIST identifies privacy, explainability, fairness, transparency, accountability and reliability as important characteristics of trustworthy AI.

FR-10: Drift monitoring

The system shall compare current data or prediction distributions against a reference period. The MVP shall support at least one of:

- Population Stability Index.
- Kolmogorov–Smirnov test.
- Chi-squared test for categorical features.

The system shall display:

- Feature drift status.
- Prediction drift status.
- Drift score.
- Reference period.
- Current period.
- Alert severity.
- Recommended investigation action.

FR-11: Audit logging

The system shall log:

- Institution registration.
- Dataset schema mapping.
- Training-round start and completion.
- Client participation.
- Privacy configuration.
- Model-version creation.
- Risk-score requests.
- Explanation generation.
- Fairness and drift evaluations.
- Configuration changes.
- Failed or abandoned rounds.

Audit entries shall include timestamp, actor or service, action, object, status and relevant model version.

FR-12: Privacy scorecard

The privacy centre shall present:

Raw records exchanged: 0
Raw columns exchanged: 0
Individual updates visible to coordinator: No
Secure aggregation: Enabled
Differential privacy: Enabled
Minimum clients required: 3
Current participating clients: 3
Privacy budget: Calculated or clearly labeled as configured
Data retention: Configured policy

REQUIREMENT NOTE

The system must not claim that federated learning alone guarantees privacy. Federated learning should be presented as one component combined with secure aggregation and differential privacy.

8. Dashboard Requirements

8.1 Overview

Show:

- Global model version.
- Current training round.
- Participating institutions.
- Global PR-AUC and recall.
- Improvement over silo-only baseline.
- Privacy status.
- Fairness status.
- Drift status.
- Recent alerts.

8.2 Institution network

Show each institution as a client node: bank, wallet, lender and optional insurer. For each institution show:

- Local row count.
- Local fraud rate.
- Local PR-AUC.
- Participation status.
- Latest update timestamp.
- Raw-data sharing status.

8.3 Live training

Show:

- Current round.
- Client training progress.

- Local model metrics.
- Protected-update transmission.
- Secure aggregation status.
- Global model metrics by round.
- Client dropout events.
- Training duration.

8.4 Model comparison

The dashboard shall compare:

Model	Purpose
Local-only	Demonstrates silo limitations
Centralized	Research upper bound only
Federated	Measures collaboration benefit
Federated + secure aggregation	Measures privacy architecture
Federated + secure aggregation + DP	Final proposed configuration

REQUIREMENT NOTE

The dashboard shall use metrics generated from actual experiments. It must not hard-code illustrative values.

8.5 Risk scoring

Show:

- Transaction input form.
- Risk score.
- Category.
- Recommended action.
- Explanation.
- Local-versus-global comparison.
- Privacy note.
- Model version.

8.6 Responsible AI centre

Show:

- Fairness metrics.
- Drift metrics.
- Data-quality alerts.
- Explanation availability.

- Human-review policy.
- Governance metadata.

9. Data Requirements

Dataset

The preferred MVP dataset is PaySim, a synthetic mobile-money transaction dataset created for fraud-detection research.

Shared schema

The shared schema may include:

Field	Field	Field
Transaction type	Time step	Device familiarity
Amount	Origin balance before transaction	New-recipient indicator
Origin balance after transaction	Destination balance before transaction	Historical transaction aggregates
Destination balance after transaction	Transaction velocity	Fraud label
Account-age proxy		

Data-silo strategy

The MVP shall use non-IID partitions to simulate institution differences. Example:

Institution	Feature view
Bank	Account and balance-heavy features.
Wallet	Transaction, device and velocity features.
Lender	Customer-history and repayment-style features.
Insurer	Optional behavioural or claims-style features.

If feature-view construction is too expensive, the team may use institution-specific non-IID partitions while clearly labeling them as a simulation.

Data quality rules

The pipeline shall:

- Remove leakage-prone identifiers.
- Preserve class-ratio metadata.
- Track missingness locally.
- Validate schema compatibility.

- Keep test data isolated.
- Record partition characteristics.
- Avoid representing ordinary hashes as complete anonymization.

10. Evaluation Plan

Primary success criteria

The MVP is successful if it demonstrates:

- Raw transaction rows shared with the coordinator: zero.
- A working multi-client federated training flow.
- Secure aggregation enabled for the final demo.
- Differential privacy enabled and visibly configured.
- Federated PR-AUC or recall improvement over the median silo-only model.
- Per-transaction explanation for a scored example.
- Fairness and drift panels populated with real experiment results.
- A complete demo runnable locally.

Model metrics

Track:

- PR-AUC.
- ROC-AUC.
- Recall.
- Precision.
- F1 score.
- False-positive rate.
- False-negative rate.
- Recall at a fixed review capacity.
- Expected fraud-control cost.

Expected cost may be represented as:

$$\text{Expected cost} = C_{\text{FN}} \times \text{FN} + C_{\text{FP}} \times \text{FP} + C_{\text{review}} \times \text{Reviews}$$

FN and FP are the false-negative and false-positive counts for the evaluation period; C_{FN} , C_{FP} and C_{review} are the configurable cost weights agreed with the fraud-control owner.

Federated metrics

Track:

- Global performance per round.
- Client-level performance per round.
- Convergence speed.

- Training duration.
- Communication volume.
- Client participation.
- Client dropout tolerance.
- Performance variation across institutions.

Required experiments

Experiment	Design
A — Utility	Compare local-only models, the federated model, the centralized research upper bound, and the federated model with privacy enhancements.
B — Privacy / utility trade-off	Compare no differential privacy, lower-noise DP and higher-noise DP. Display the impact on PR-AUC, recall, false-positive rate, privacy budget and training stability.
C — Robustness	Simulate one client dropping out, unequal client sizes, different fraud rates, a low-quality client, a delayed client and non-IID feature distributions.

11. Technical Architecture

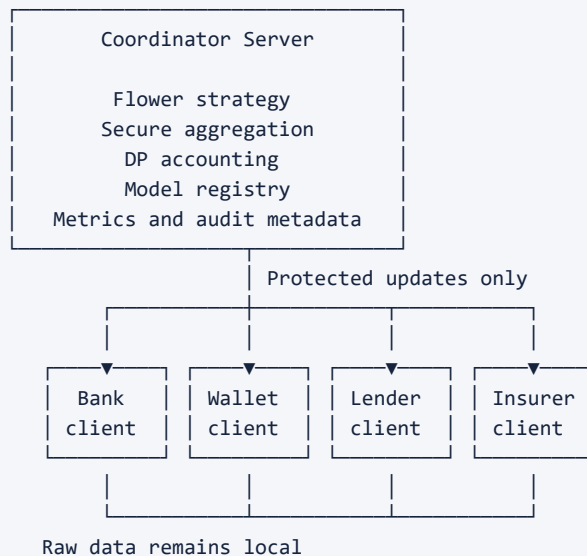


Figure 1 — TrustFed reference architecture: raw data remains inside each institution; only protected updates reach the coordinator.

Suggested stack

Layer	Technology
Federated learning	Flower
Model training	PyTorch

Layer	Technology
Differential privacy	Flower workflow and/or Opacus
Explanations	SHAP
Backend	FastAPI
Dashboard	Streamlit for MVP; React and TypeScript for polished build
Charts	Plotly, Recharts or Streamlit charts
Metadata	SQLite or PostgreSQL
Experiment tracking	MLflow
Deployment	Docker Compose
Live updates	WebSockets or polling

REQUIREMENT NOTE

Flower documents federated evaluation, secure aggregation and differential privacy examples that can accelerate the prototype.

12. API Requirements

Start training

POST /training/start

Request:

```
{
  "selected_clients": ["bank", "wallet", "lender"],
  "rounds": 10,
  "local_epochs": 2,
  "privacy_mode": "secure_aggregation_dp",
  "minimum_clients": 3
}
```

Response:

```
{
  "job_id": "job_001",
  "status": "started"
}
```

Training status

GET /training/{job_id}

Response:

```
{
  "job_id": "job_001",
  "round": 4,
  "total_rounds": 10,
  "status": "running",
  "participating_clients": 3,
  "global_pr_auc": 0.53,
  "secure_aggregation": true,
  "differential_privacy": true
}
```

Score transaction

POST /risk/score

Request:

```
{
  "transaction": {
    "amount": 12500,
    "transaction_type": "TRANSFER",
    "new_recipient": true,
    "device_seen_before": false,
    "transfers_last_90_seconds": 4
  },
  "model_version": "global_v10"
}
```

Response:

```
{
  "risk_score": 87,
  "fraud_probability": 0.91,
  "risk_category": "high",
  "recommended_action": "manual_review",
  "model_version": "global_v10",
  "reasons": [
    "Amount is above the customer baseline",
    "Destination account is new",
    "Device has not been seen previously",
    "Rapid repeated transfers"
  ]
}
```

13. Threat Model and Safety Requirements

Assumptions

- The coordinator may be honest-but-curious.
- Institutions do not share raw records.
- Clients are simulated and generally cooperative.
- Secure aggregation protects individual updates from the coordinator.

- Differential privacy limits information leakage from model contributions.

Out of scope for MVP

- Colluding clients.
- Poisoning attacks.
- Sybil clients.
- Backdoor attacks.
- Full secure hardware deployment.
- Formal cryptographic security review.

Safety requirements

1. Never expose raw client records in coordinator logs.
2. Do not display individual client updates.
3. Enforce a minimum participation threshold.
4. Label simulated privacy values clearly.
5. Do not claim legal compliance automatically.
6. Provide human review for high-risk decisions.
7. Log model version and explanation for every risk score.
8. Display uncertainty or insufficient-data warnings where appropriate.

14. Governance Requirements

The governance screen shall include:

- Processing purpose.
- Data categories used.
- Institution policy.
- Consent or lawful-basis metadata for simulation.
- Retention period.
- Model version.
- Explanation availability.
- Opt-out or deletion workflow status.
- Audit-log access.

REQUIREMENT NOTE

The product should state that privacy-preserving architecture does not by itself establish compliance with any particular law. Compliance depends on the complete processing design, contracts, notices, security controls, retention practices and deployment context.

15. Demo Flow

Time	Step	Content
Minute 1	Problem	Show three institutional silos: bank (local transaction records), wallet (device and mobile-money records), lender (repayment and customer-history records). Explain that raw records remain at each institution.
Minute 2	Federated training	Click Start training and show: clients train locally, raw data remains inside each silo, updates are clipped and protected, secure aggregation combines updates, the global model improves by round, and privacy and participation status update live.
Minute 3	Risk decision	Submit a suspicious transaction and show: fraud probability, risk score, manual-review recommendation, four explanation factors, local-versus-global comparison, fairness status, drift status and the privacy scorecard.

REQUIREMENT NOTE

"TrustFed enables institutions to share model intelligence without sharing customer records."

16. Delivery Plan

Phase	Focus	Deliverables
Phase 1	Baseline	Load and clean PaySim; build time-aware splits; train a local model; implement metrics; add basic risk-scoring API; add explanations.
Phase 2	Federated learning	Create three simulated clients; implement Flower simulation; add FedAvg; compare local and federated models; record metrics per round.
Phase 3	Privacy	Add secure aggregation; add minimum-client threshold; add differential privacy; add privacy accounting and dashboard state; validate that raw records never reach the coordinator.
Phase 4	Responsible AI	Add fairness metrics; add drift monitoring; add human-review policy; add governance and audit screens.
Phase 5	Product polish	Add live training progress; add model-comparison charts; add privacy flow visualization; add demo transaction; package with Docker Compose; prepare reproducible demo data and results.

17. Acceptance Criteria

The MVP is accepted when:

1. Three simulated institutions can join a training round.
2. Each client preprocesses and trains locally.
3. Raw records are not transmitted to the coordinator.
4. The coordinator cannot inspect an individual update in the final configured flow.
5. The system blocks aggregation below the minimum client threshold.
6. Differential-privacy settings are visible and logged.

7. Local and federated metrics are generated from real experiments.
8. A global model can score a transaction.
9. The score includes an explanation and recommended human-review action.
10. Fairness metrics are visible for at least two groups.
11. Drift monitoring displays a real or simulated comparison.
12. Audit logs capture training and scoring events.
13. The complete demonstration runs end to end on one machine.

18. Final Product Positioning

TrustFed is a federated financial-risk intelligence platform for institutions that need better fraud detection without centralizing customer data.

Its differentiation is not federated learning alone. The product combines:

Capability	What it delivers
Collaborative model training	Joint fraud-risk model across bank, wallet and lender silos.
Secure aggregation	Coordinator cannot inspect individual client updates.
Differential privacy	Quantified privacy budget with clipping, noise and accounting.
Explainable risk decisions	Per-transaction reason codes and recommended actions.
Fairness monitoring	Group-level recall, FPR and equal-opportunity reporting.
Drift monitoring	Feature and prediction drift against a reference period.
Governance and auditability	Audit log, privacy scorecard and governance metadata.
Measurable comparison	A local-versus-global performance comparison from real runs.

REQUIREMENT NOTE

Institutions collaborate on intelligence, not customer data.